

Prediction Does Not Imply Control: Geometry and Random-Matrix Theory of Self-Generated Distribution Shift

Author list to be added

Abstract

- Predictive models of neural responses can have similar held-out accuracy yet differ sharply in their ability to synthesize stimuli that control the true response.
- We formalize this dissociation in a noisy linear teacher–student model. Natural generalization is measured in the data-covariance geometry, whereas gradient-based accentuation is evaluated along a direction chosen by the fitted model itself.
- Random-matrix deterministic equivalents quantify generalization, accentuation, peer review, and cross-validated ridge selection under anisotropic data.
- For regression after a fixed linear feature map, prediction depends on a smaller set of summary statistics than pixel-space control. We construct continuous families of feature maps with identical prediction and cross-validation behavior but dramatically different accentuation.
- Natural-image covariance experiments validate the theory and show that moving prediction-equivalent features into low-variance spectral directions can leave prediction intact while driving accentuation R^2 from near one to very negative values.

Status and intended scope

- This document is an author-facing manuscript skeleton, not a complete draft.
- Bullets specify the intended logical content of each paragraph.
- Formal statements identify the results that should survive into the main text. Proofs and long deterministic-equivalent calculations will be placed in the supplement.
- Citations, biological details, author list, and journal formatting are placeholders.

1 Introduction: the prediction–control paradox

- Start from the empirical observation: encoding models with comparable held-out prediction can generate qualitatively different preferred images and can differ sharply in closed-loop control.
- Highlight the evaluation asymmetry: a poor generator can sometimes evaluate stimuli produced by a better generator even though its own synthesized stimuli fail under the ground-truth response.

Figure 1 placeholder: empirical paradox and minimal geometry. Left: representative natural-image prediction, successful versus failed accentuation, and peer-review asymmetry. Right: natural inputs probe the covariance ellipsoid, whereas accentuation moves along the fitted gradient.

Figure 1: Similar prediction does not guarantee similar control because the model changes the distribution on which it is evaluated.

- State the conceptual reason. Standard generalization evaluates a predictor on an externally specified natural distribution. Control evaluates it on a distribution selected by the fitted predictor itself.
- State the paper’s thesis: prediction and gradient-based control depend on different geometries and, in feature models, on different identifiable summary statistics.
- Explain the role of random-matrix theory: the geometric gap is exact, while RMT predicts how finite samples, response noise, anisotropy, ridge shrinkage, and cross-validation determine its magnitude.
- Preview the strongest result: a continuous family of feature maps can have identical Gaussian training distributions and identical prediction-optimal ridge penalties, while its pixel-space control changes by orders of magnitude.

2 Teacher–student setup and evaluation metrics

- Let centered inputs and noisy responses follow

$$\mathbf{x} \sim \mathcal{N}(0, \Sigma), \quad y = (\boldsymbol{\beta}^*)^\top \mathbf{x} + \sigma \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1). \quad (1)$$

- Define the noiseless signal variance

$$S = (\boldsymbol{\beta}^*)^\top \Sigma \boldsymbol{\beta}^*. \quad (2)$$

- Distinguish evaluation against the noiseless teacher from evaluation against a newly observed noisy response. The main text uses the noiseless teacher; test noise can be added as an irreducible term.
- For a fixed fitted pixel-space weight $\hat{\boldsymbol{\beta}}$, define natural generalization and own-gradient accentuation from the same underlying teacher.

Proposition 1 (Exact prediction and own-path control metrics). *For a fixed fitted weight $\hat{\boldsymbol{\beta}}$, natural generalization satisfies*

$$E_{\text{gen}} = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)^\top \Sigma (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*), \quad (3)$$

$$R_{\text{gen}}^2 = 1 - \frac{E_{\text{gen}}}{S}. \quad (4)$$

Along the self-generated path $\mathbf{x}(t) = \mathbf{x}_0 + t\hat{\boldsymbol{\beta}}$, define

$$N_{\text{acc}} = \hat{\boldsymbol{\beta}}^\top \boldsymbol{\beta}^*, \quad D_{\text{acc}} = \hat{\boldsymbol{\beta}}^\top \hat{\boldsymbol{\beta}}. \quad (5)$$

The true-on-predicted path slope, student-range-normalized accentuation loss, and true-path coefficient of determination are

$$b_{\text{acc}} = \frac{N_{\text{acc}}}{D_{\text{acc}}}, \quad (6)$$

$$\frac{E_{\text{acc}}}{S} = \left(1 - \frac{N_{\text{acc}}}{D_{\text{acc}}}\right)^2, \quad (7)$$

$$R_{\text{acc}}^2 = 1 - \left(1 - \frac{D_{\text{acc}}}{N_{\text{acc}}}\right)^2. \quad (8)$$

- Explain the two opposite singularities. E_{acc}/S is unstable when the fitted gradient norm D_{acc} approaches zero; R_{acc}^2 is unstable when the true path gain N_{acc} approaches zero.
- Emphasize that a very negative R_{acc}^2 need not mean an unbounded raw pathwise MSE: when $N_{\text{acc}} \rightarrow 0$ with $D_{\text{acc}} > 0$, $R_{\text{acc}}^2 \rightarrow -\infty$ while $E_{\text{acc}}/S \rightarrow 1$.
- Use b_{acc} for slope and $z_{\text{acc}} = D_{\text{acc}}/N_{\text{acc}} - 1$ for the R^2 -controlling ratio. Do not call either quantity R_{det} .

3 Direct pixel regression: anisotropy, noise, and ridge

- Fit normalized ridge regression using n samples:

$$\hat{\boldsymbol{\beta}} = (\hat{\Sigma} + \lambda I)^{-1} \left(\hat{\Sigma} \boldsymbol{\beta}^* + \frac{\sigma}{n} X^\top \boldsymbol{\eta} \right), \quad \hat{\Sigma} = \frac{X^\top X}{n}. \quad (9)$$

- Make the ridge convention visible: software minimizing $\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \alpha \|\boldsymbol{\beta}\|^2$ uses $\alpha = n\lambda$.
- Introduce the effective deterministic-equivalent penalty κ :

$$\kappa - \lambda = \frac{\kappa}{n} \text{tr}[\Sigma(\Sigma + \kappa I)^{-1}]. \quad (10)$$

- Define compact spectral summaries

$$B_{p,q}(\kappa) = (\boldsymbol{\beta}^*)^\top \Sigma^p (\Sigma + \kappa I)^{-q} \boldsymbol{\beta}^*, \quad (11)$$

$$\text{df}_{p,q}(\kappa) = \text{tr}[\Sigma^p (\Sigma + \kappa I)^{-q}], \quad (12)$$

with $\text{df}_2 = \text{df}_{2,2}$.

Proposition 2 (Pixel-space deterministic equivalents). *In the proportional high-dimensional limit, the leading deterministic equivalent of noiseless generalization error is*

$$E_{\text{gen}}^{\text{DE}} = \frac{n}{n - \text{df}_2} (\kappa^2 B_{1,2} + \sigma^2) - \sigma^2. \quad (13)$$

The leading means of the accentuation numerator and denominator are

$$\mu_N = B_{1,1}, \quad (14)$$

$$\mu_D = B_{2,2} + \frac{\text{df}_{1,2}}{n - \text{df}_2} (\kappa^2 B_{1,2} + \sigma^2) \quad (15)$$

$$= B_{2,2} + \frac{\text{df}_{1,2}}{n} (E_{\text{gen}}^{\text{DE}} + \sigma^2). \quad (16)$$

Figure 2 placeholder: pixel-ridge mechanism and RMT validation. (a) Data spectrum and teacher power. (b) Mean per-PC coefficient error, emphasizing low-variance directions. (c) Noise sweep with DE and Monte Carlo for R_{gen}^2 and R_{acc}^2 . (d) DE-CV and empirical CV ridge penalties. Optional peer-review panel.

Figure 2: RMT predicts the prediction–accentuation gap after cross-validated ridge selection.

Consequently, the typical-fit leading approximation is

$$z_{\text{acc}}^{\text{DE}} = \frac{\mu_D}{\mu_N} - 1 = \frac{\sigma^2 \text{df}_{1,2} - n\lambda B_{1,2}}{(n - \text{df}_2) B_{1,1}}. \quad (17)$$

- Interpret the per-PC error decomposition in one paragraph or inset: ridge bias suppresses teacher signal below the κ scale, while finite-sample noise is largest in low-variance modes near that scale.
- Explain the central dissociation: E_{gen} weights coefficient error by Σ , whereas D_{acc} measures unweighted pixel gradient energy.
- Keep the one- and two-resolvent derivations in the supplement.

Proposition 3 (Prediction-optimal ridge need not optimize control). *At an interior stationary point of the DE generalization curve,*

$$(n - \text{df}_2) \kappa B_{2,3} = \text{df}_{2,3} (\kappa^2 B_{1,2} + \sigma^2). \quad (18)$$

Substitution into Eq. (??) gives

$$z_{\text{acc,CV}}^{\text{DE}} = \kappa \frac{\text{df}_{1,2}}{B_{1,1}} \left(\frac{B_{2,3}}{\text{df}_{2,3}} - \frac{B_{1,2}}{\text{df}_{1,2}} \right). \quad (19)$$

The right-hand side is not generically zero. Thus minimizing natural prediction risk does not in general calibrate the self-generated path.

- Separate the population/DE-CV optimum in Proposition ?? from random fold-to-fold selection of α . The latter is a finite-sample fluctuation analyzed in the supplement.
- Use a dense logarithmic α grid in all experiments so apparent transitions are not artifacts of coarse regularization stages.

4 Linear features create two geometries

- Map pixels to p fixed linear features and fit ridge in feature coordinates:

$$\mathbf{z} = F\mathbf{x}, \quad \hat{\mathbf{f}}(\mathbf{x}) = \hat{\boldsymbol{\theta}}^\top F\mathbf{x}, \quad \hat{\boldsymbol{\beta}} = F^\top \hat{\boldsymbol{\theta}}. \quad (20)$$

- Define the prediction summaries

$$\mathbf{C} = F\Sigma F^\top, \quad \mathbf{h} = F\Sigma\boldsymbol{\beta}^*, \quad (21)$$

and the pixel-control summaries

$$\mathbf{G} = FF^\top, \quad \mathbf{q} = F\boldsymbol{\beta}^*. \quad (22)$$

- Explain the distinction: C is the natural feature covariance, while G measures the pixel norm of a back-propagated feature-space direction. Likewise, $\mathbf{h}$ is the feature–teacher covariance on natural inputs, while $\mathbf{q}$ is the teacher gain along a pixel gradient.

Proposition 4 (Prediction-optimal and control-optimal feature weights). *The best population predictor in the feature span and its irreducible error are*

$$\boldsymbol{\theta}^* = C^\dagger \mathbf{h}, \quad (23)$$

$$S_\perp = S - \mathbf{h}^\top C^\dagger \mathbf{h}. \quad (24)$$

The feature coefficient whose back-projected pixel weight is closest to the teacher in Euclidean norm is

$$\boldsymbol{\theta}_E = G^\dagger \mathbf{q}. \quad (25)$$

Prediction training centers the estimator at $\boldsymbol{\theta}^$ in the C geometry, whereas exact pixel-gradient control is centered at $\boldsymbol{\theta}_E$ in the G geometry. These centers need not agree.*

Proposition 5 (Feature-space deterministic equivalents). *Let κ solve Eq. (??) with Σ replaced by C , and define*

$$\mathcal{R} = (C + \kappa I)^{-1}, \quad \boldsymbol{\mu} = C\mathcal{R}\boldsymbol{\theta}^*, \quad (26)$$

$$v_0 = \frac{\kappa^2(\boldsymbol{\theta}^*)^\top \mathcal{R} C \mathcal{R} \boldsymbol{\theta}^* + S_\perp + \sigma^2}{n - \text{tr}(C^2 \mathcal{R}^2)}. \quad (27)$$

Then

$$E_{\text{gen}}^{\text{DE}} = S_\perp + (\boldsymbol{\mu} - \boldsymbol{\theta}^*)^\top C(\boldsymbol{\mu} - \boldsymbol{\theta}^*) + v_0 \text{tr}(C\mathcal{R}C\mathcal{R}), \quad (28)$$

$$N_{\text{acc}}^{\text{DE}} = \mathbf{q}^\top \boldsymbol{\mu}, \quad (29)$$

$$D_{\text{acc}}^{\text{DE}} = \boldsymbol{\mu}^\top G \boldsymbol{\mu} + v_0 \text{tr}(G\mathcal{R}C\mathcal{R}). \quad (30)$$

Equivalently, $v_0 = (E_{\text{gen}}^{\text{DE}} + \sigma^2)/n$.

Theorem 6 (Prediction statistics do not identify pixel control). *For Gaussian inputs and a deterministic feature map, the joint distribution of the training features and responses is determined by*

$$(C, \mathbf{h}, S, \sigma^2). \quad (31)$$

Consequently, the distribution of the fitted ridge coefficient, the complete natural generalization curve, and population/DE cross-validated ridge selection depend only on

$$(C, \mathbf{h}, S, \sigma^2, n). \quad (32)$$

Pixel-gradient accentuation additionally depends on

$$(G, \mathbf{q}) = (FF^\top, F\boldsymbol{\beta}^*). \quad (33)$$

Therefore prediction and cross-validation alone do not identify E_{acc} , b_{acc} , or R_{acc}^2 .

- Present the summary-statistics dependency graph here, immediately after Theorem ??.
- For non-Gaussian natural inputs, state the weaker exact claim: $(C, \mathbf{h}, S)$ determine the second-order prediction theory, but need not determine the full feature–response distribution.

Figure 3 placeholder: dependency graph. Primitive quantities (F, Σ, β^*) feed into a prediction branch $(C, \mathbf{h}, S, \theta^*, S_\perp, \kappa_{CV}, E_{\text{gen}})$ and a control branch that additionally requires $(G, \mathbf{q})$. Keep the two branches visually separated and left-to-right topologically ordered.

Figure 3: Prediction and pixel control use overlapping but nonidentical summary statistics.

5 An exact prediction-preserving family with changing control

- Move to whitened input coordinates:

$$A = F\Sigma^{1/2}, \quad \mathbf{b} = \Sigma^{1/2}\beta^*, \quad \mathbf{c} = \Sigma^{-1/2}\beta^*. \quad (34)$$

- In these coordinates, $C = AA^\top$, $\mathbf{h} = A\mathbf{b}$, $\mathbf{q} = A\mathbf{c}$, while $G = A\Sigma^{-1}A^\top$.

Theorem 7 (Prediction-preserving gauge transformation). *Let $Q \in O(d)$ be orthogonal and satisfy $Q\mathbf{b} = \mathbf{b}$. Define*

$$F_Q = F\Sigma^{1/2}Q\Sigma^{-1/2} = AQ\Sigma^{-1/2}. \quad (35)$$

Then

$$F_Q\Sigma F_Q^\top = F\Sigma F^\top = C, \quad F_Q\Sigma\beta^* = F\Sigma\beta^* = \mathbf{h}. \quad (36)$$

Hence the Gaussian training distribution, ridge estimator in feature coordinates, natural generalization curve, and cross-validated ridge penalty are invariant under Q . In contrast,

$$\mathbf{q}_Q = F_Q\beta^* = AQ\mathbf{c}, \quad (37)$$

$$G_Q = F_QF_Q^\top = AQ\Sigma^{-1}Q^\top A^\top, \quad (38)$$

which need not be invariant.

Corollary 8 (Strong gauge: isolate the back-propagation metric). *If Q fixes both $\mathbf{b}$ and $\mathbf{c}$, then C , $\mathbf{h}$, and $\mathbf{q}$ are all invariant, while G_Q can still change whenever Σ is anisotropic. Along this stronger family, the distribution of $N_{\text{acc}} = \mathbf{q}^\top \hat{\boldsymbol{\theta}}$ is fixed and all changes in accentuation arise through the back-propagated gradient energy $D_{\text{acc}} = \hat{\boldsymbol{\theta}}^\top G_Q \hat{\boldsymbol{\theta}}$.*

- Explain why the construction sometimes projects away the span of both $\mathbf{b}$ and $\mathbf{c}$. Fixing $\mathbf{b}$ alone is sufficient to preserve prediction. Fixing $\mathbf{c}$ as well preserves the accentuation numerator and yields the cleanest causal dissection of G .
- Connect the gauge to anisotropy. If $\Sigma = I$, then $G_Q = AA^\top$ is invariant and the nontrivial dissociation disappears.

5.1 Continuous top-PC-to-tail path used in the experiment

- Use the strong-gauge subspace so every moving component annihilates both $\mathbf{b}$ and $\mathbf{c}$.
- Decompose the whitened feature map into a fixed part P , a well-behaved top-PC component R , and a matched tail component V :

$$A(t) = P + \sqrt{1-t}R + \sqrt{t}V, \quad F(t) = A(t)\Sigma^{-1/2}, \quad 0 \leq t \leq 1. \quad (39)$$

Figure 4 placeholder: main constructive result. Use the Van Hateren $p = 500$ top-PC-to-tail rotation. Top row: $\text{tr}(FF^\top)$, invariant E_{gen} , and invariant R_{gen}^2 . Bottom row: E_{acc}/S , R_{acc}^2 , and b_{acc} across $t = \sin^2 \theta$, with noise levels encoded by color. Pair this with selected feature-row spectra at $t = 0, 10^{-4}, 10^{-2}, 1$.

Figure 4: A prediction-equivalent feature rotation moves from a controlled top-PC endpoint to catastrophic pixel-space accentuation.

- State the construction constraints:

$$R\mathbf{b} = V\mathbf{b} = 0, \quad R\mathbf{c} = V\mathbf{c} = 0, \quad RR^\top = VV^\top, \quad (40)$$

together with the required cross-orthogonality to P and between the rotating components. These conditions make $A(t)A(t)^\top$, $A(t)\mathbf{b}$, and $A(t)\mathbf{c}$ invariant.

- Parameterize $t = \sin^2 \theta$, so that $\sqrt{1-t} = \cos \theta$ and $\sqrt{t} = \sin \theta$. The path is an ordinary continuous rotation in whitened coordinates.
- Choose $t = 0$ to be the unwhitened top- p PC projector. Choose V in very low-variance spectral directions, making $G(t)$ grow while prediction remains fixed.

Proposition 9 (Protected prediction can approach singular control). *Assume C is positive definite on the effective feature support. The smallest excess natural-prediction loss among feature weights with zero true accentuation gain is*

$$\epsilon_{N=0} = \min_{\boldsymbol{\theta}: \mathbf{q}^\top \boldsymbol{\theta} = 0} \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_C^2 = \frac{(\mathbf{q}^\top \boldsymbol{\theta}^*)^2}{\mathbf{q}^\top C^{-1} \mathbf{q}}. \quad (41)$$

Thus nearly perfect natural prediction can lie arbitrarily close to the R_{acc}^2 singularity when both $S_\perp/S$ and $\epsilon_{N=0}/S$ are small. The susceptibility $\mathbf{q}^\top C^{-1} \mathbf{q}$ is large when the control functional points into low-variance feature directions.

6 Natural-image experiments

6.1 Measured covariance and disk teacher

- Introduce the Van Hateren and FFHQ image ensembles, patch/image dimensions, covariance estimation, retained rank, and circular disk teacher.
- Plot the image spectrum and teacher power in the population-PC basis.
- Report response noise on two aligned axes: response standard deviation σ and noise-to-signal variance ratio σ^2/S .
- Explain which experiments draw Gaussian samples with the measured covariance and which use actual natural-image samples. Do not imply exact Gaussian distributional theory for the latter.

Figure 5 placeholder: representation comparison. Noise sweeps for full PCA, whitening, and top-PC truncation. Overlay R_{gen}^2 and R_{acc}^2 in one panel, and b_{gen} and b_{acc} in another. Include the top-PC cutoff sweep if space permits.

Figure 5: Feature conditioning for prediction can worsen pixel-space control; spectral truncation can reverse the gap.

6.2 Cross-validated pixel ridge

- Compare fixed ridge and prediction-selected RidgeCV across noise.
- Show that CV protects generalization substantially but does not make accentuation an optimized objective.
- Plot the selected α path beneath vertically aligned panels for E , R^2 , and slope so transitions can be related to regularization.
- Use the leading DE for typical values and the second-order or distributional treatment for nonlinear ratio averages.

6.3 PCA, whitening, and top-PC truncation

Feature map	feature loading	C spectrum	G spectrum	omitted signal
Full PCA	1	s_j	1	0
Full whitening	$s_j^{-1/2}$	1	s_j^{-1}	0
Top- K PCs	$1, j \leq K$	s_j	1	$\sum_{j>K} s_j b_j^2$

- Full PCA is the direct pixel baseline in rotated coordinates.
- Whitening makes regression isotropic in feature space but amplifies low-variance directions by $1/s_j$ when gradients are mapped back to pixels.
- Top-PC truncation pays an irreducible prediction cost but removes harmful low-variance control directions. When teacher signal is concentrated in top PCs, this trade can improve accentuation greatly at modest predictive cost.

7 Finite-size fluctuations of nonlinear control metrics

- Explain why a ratio of deterministic means can predict typical accentuation while failing for an empirical mean near a singular denominator.
- Retain an approximate Gaussian distribution for the fitted feature coefficient:

$$\hat{\theta} \sim \mathcal{N}(\mu, V), \quad V = v_0 R C R. \quad (42)$$

Proposition 10 (Gaussian distributional deterministic equivalent). *Under Eq. (??), the linear-quadratic accentuation moments are*

$$\mathbb{E}N_{\text{acc}} = \mathbf{q}^\top \boldsymbol{\mu}, \quad \text{Var}(N_{\text{acc}}) = \mathbf{q}^\top V \mathbf{q}, \quad (43)$$

$$\mathbb{E}D_{\text{acc}} = \boldsymbol{\mu}^\top G \boldsymbol{\mu} + \text{tr}(GV), \quad \text{Var}(D_{\text{acc}}) = 2 \text{tr}[(GV)^2] + 4 \boldsymbol{\mu}^\top GVG \boldsymbol{\mu}, \quad (44)$$

$$\text{Cov}(N_{\text{acc}}, D_{\text{acc}}) = 2 \mathbf{q}^\top VG \boldsymbol{\mu}. \quad (45)$$

Sampling $\hat{\boldsymbol{\theta}}$ from this surrogate and applying the exact per-fit metric definitions yields approximate means, medians, intervals, and tail probabilities without replacing random ratios by ratios of means.

- Diagnose denominator concentration using

$$\text{CV}_N = \frac{\sqrt{\text{Var}(N_{\text{acc}})}}{|\mathbb{E}N_{\text{acc}}|}, \quad \text{CV}_D = \frac{\sqrt{\text{Var}(D_{\text{acc}})}}{\mathbb{E}D_{\text{acc}}}. \quad (46)$$

- Distinguish two questions: whether the leading DE is statistically adequate, and whether control itself is good. Increasing p can make the leading DE more accurate while making accentuation worse.
- Keep the full random-design $\text{Var}(R)$ and CV-selection fluctuation theory in the supplement. In the main paper, report medians/IQRs as well as means/standard errors near ratio singularities.

8 Peer review and cross-model asymmetry

- Let generator and evaluator have pixel weights $\hat{\boldsymbol{\beta}}_g$ and $\hat{\boldsymbol{\beta}}_e$. Along the generator path, define the teacher and evaluator gains

$$T_g = (\boldsymbol{\beta}^\star)^\top \hat{\boldsymbol{\beta}}_g, \quad D_{e \leftarrow g} = \hat{\boldsymbol{\beta}}_e^\top \hat{\boldsymbol{\beta}}_g. \quad (47)$$

- The peer-review coefficient of determination is

$$R_{e \leftarrow g}^2 = 1 - \left(1 - \frac{D_{e \leftarrow g}}{T_g}\right)^2. \quad (48)$$

- Explain the asymmetry geometrically: one generator can move along a direction that remains legible to a good evaluator, while the reverse direction has nearly zero teacher gain.
- Treat peer review as a consequence and validation of the prediction-control geometry, not as a second independent theoretical spine.
- Place the full joint-resolvent theory for correlated fits in the supplement or future work.

9 Discussion

- Main conclusion: predictive equivalence is weaker than control equivalence because intervention exposes geometry not identifiable from natural prediction alone.
- Cross-validation solves the objective it observes. It can select the prediction-optimal ridge penalty while leaving substantial or catastrophic control error.

- Low-variance directions provide a shared mechanism for the pixel and feature cases, but feature maps add an irreducible structural mismatch between prediction-optimal and control-optimal projections.
- Whitening illustrates that a representation can be statistically well conditioned for regression while being ill conditioned for pixel-space intervention.
- Top-PC truncation illustrates a practical design principle: discard directions that contribute little target signal but large intervention gain.
- Relate the linear theory cautiously to nonlinear neural networks: local input gradients define an input-space control metric, while representation covariance governs prediction on natural data.
- State limitations: Gaussian or second-order input model, fixed linear features, linear teacher, local gradient paths, and incomplete unconditional fluctuation theory for nonlinear ratios and random CV.

Proposed supplementary organization

1. Exact generalization, accentuation, and peer-review metric derivations.
2. One- and two-resolvent deterministic-equivalent identities.
3. Per-PC bias and variance decomposition for direct ridge regression.
4. Derivation of the CV stationarity condition and Eq. (??).
5. Arbitrary fixed-feature-map DE and aligned spectral specialization.
6. Proofs of Theorems ?? and ??, including the continuous rotation construction.
7. Delta-method corrections, conditional $\text{Var}(R)$, missing random-design terms, and CV-selection variability.
8. Gaussian distributional DE and effective-rank diagnostics.
9. Additional dimensions, random null-space rotations, FFHQ replication, and sensitivity analyses.

Open author decisions before full drafting

- Choose whether the primary empirical opening is the original neural control result or the natural-image disk-teacher demonstration.
- Choose one primary natural-image ensemble for the main text; place the second ensemble in the supplement as replication.
- Decide whether peer review receives a main figure panel or only a concise corollary plus supplementary validation.
- Decide whether the main paper reports both E_{acc}/S and R_{acc}^2 throughout. Keeping both is theoretically useful because they expose opposite ratio singularities.

- Decide how much of the Gaussian distributional DE belongs in the main Results versus Methods/Supplement.
- Select a target venue and length before converting the bullets to complete prose.